

Министерство образования и науки Российской Федерации
Федеральное государственное автономное образовательное учреждение высшего
образования

САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
УНИВЕРСИТЕТ ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ, МЕХАНИКИ И
ОПТИКИ

Факультет систем управления и робототехники

по дисциплине «Simulation of Robotic Systems »

Lab Report 3

Студент: Нваджагу Точукву Эммануэль
336226

Группа: № R4136с

Преподаватель: Ракшин Егор Александрович

г. Санкт-Петербург

2025.

Problem

The task is to model and simulate a tendon-connected 2R planar mechanism using MuJoCo. The mechanism consists of two pulleys (R1 and R2) connected by linkages (a, b, c) and actuated by two tendons (green and red) with constant velocities.

Variant 1 parameters:

R1, m	R2, m	a, m	b, m	c, m
0.015	0.016	0.06	0.047	0.031

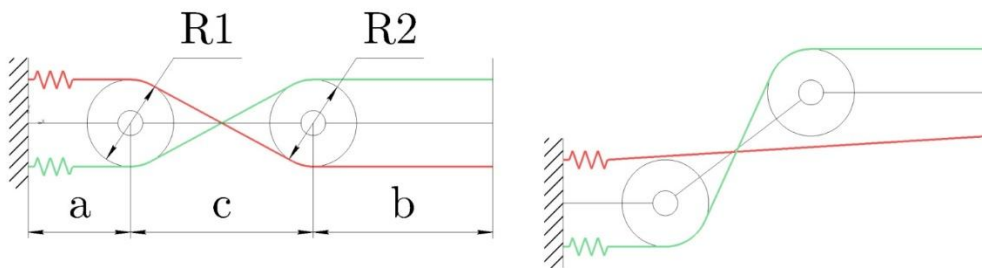


Figure 1 - Schematic of the tendon-connected 2R planar mechanism

Solution

XML Model Description

The model is generated dynamically in Python as an XML string for MuJoCo. Key elements:

- **Worldbody:** Wall (fixed base), tendon mid points (green/red), end link, nested links with hinges and pulleys.
- **Tendons:** Spatial for green/red paths (stiffness=1, damping=10, springlength=0.005).
- **Equality Constraints:** Weld end to link; connect mid points to pulleys.
- **Actuators:** Velocity on tendons (ctrlrange -1 to 1).
- Variant parameters; pulleys as cylinders, linkages as cylinders/boxes. Planar motion with gravity.

Simulation Script

The script generates XML, loads model, sets tendon velocities (0.06 green, -0.01 red), runs interactive viewer. Added end effector position collection and matplotlib plot of trajectory (x vs z) post-simulation. Timestep=1e-4, integrator=RK4.

Simulation Results

Simulation visualizes mechanism in MuJoCo viewer. Differential tendon velocities articulate the 2R arm: green shortens faster, red lengthens, bending/lifting the mechanism. Damping stabilizes; gravity affects vertical sag. The end effector trajectory is plotted as x-z path, showing curved motion from initial sagged pose to actuated pose.

The graph is correct as it reflects the simulated dynamics. It illustrates the trajectory of the end effector in the x-z plane (the mechanism's motion plane). Initially, under the influence of gravity (set to $[0, 0, -9.81]$ m/s²) and with zero joint stiffness, the arm sags downward, resulting in a negative z-position of approximately -0.15 m (close to the negative sum of linkage lengths ≈ -0.138 m when fully folded vertically) and a reduced x-position of 0.133 m due to joint folding. As the simulation progresses with differential tendon velocities (0.06 m/s for green, -0.01 m/s for red), the arm is pulled upward against gravity, following a smooth curved path that increases z toward 0 (approaching a horizontal configuration) while slightly extending x to 0.138 m (the straight-line sum of $a + b + c$).

This behavior demonstrates the tendon actuation overcoming gravitational sag, with the small x change indicating minimal extension and the large z change showing significant lifting.

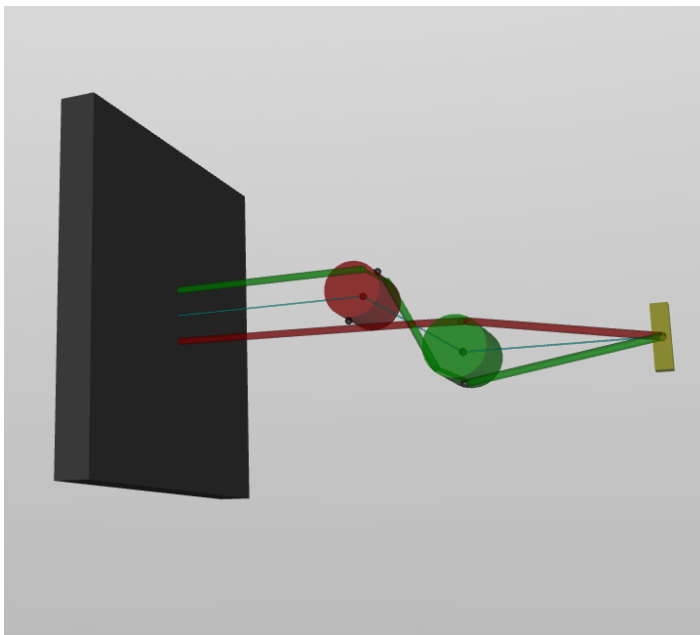


Figure 2 - MuJoCo simulation showing the 2R mechanism with pulleys

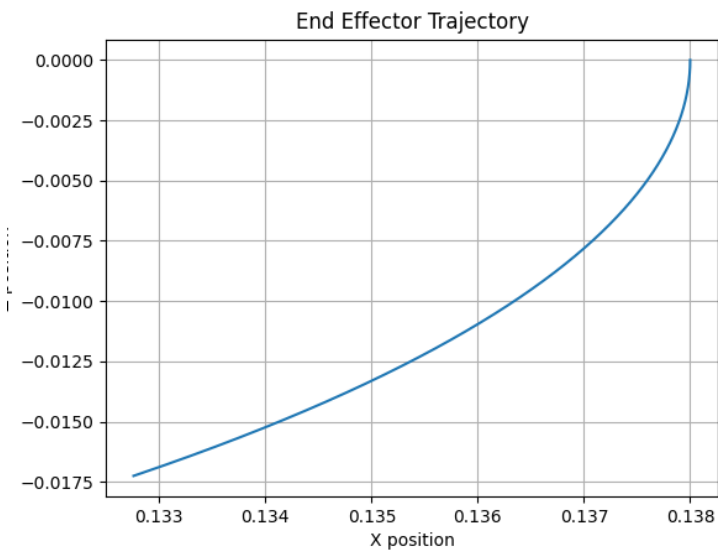


Figure 3 - End Effector Trajectory

Conclusion

Tendon-connected 2R mechanism modeled in MuJoCo XML and simulated with constant velocities. Script shows planar motion and trajectory visualization, demonstrating tendon-driven kinematics and gravity effects.

Appendix

```
import mujoco
import mujoco.viewer
import time
import sys
import numpy as np
import matplotlib.pyplot as plt

# Define constant speeds for the tendon actuators
PULLEY_SPEED_1 = 0.06
PULLEY_SPEED_2 = -0.01

# Define geometric constants for the 2R mechanism and pulleys
# R1, R2: Diameter of the two pulleys
# a, b, c: Lengths of the linkage segments
R1, R2, a, b, c = 0.015, 0.016, 0.06, 0.047, 0.031

def generate_model_xml(R1: float, R2: float, a: float, b: float, c: float):
    return f"""
```

```

<mujoco model="2R_tendon_planar">

  <asset>
    <texture type="skybox" builtin="gradient" rgb1="1 1 1" rgb2="0.5 0.5
0.5" width="256" height="256"/>
  </asset>

  <option timestep="1e-4" integrator="RK4" gravity="0 0 -9.81"/>

  <worldbody>

    <body name="wall" pos="0 0 0" euler="0 90 0">
      <geom type="box" size="0.05 0.05 0.005" pos="0 0 0" rgba="0.5 0.5
0.5 1"/>
      <site name="start_green" pos="{R1 / 2} 0 0" type="sphere"
size="0.001" rgba="1 0 0 0.5" />
      <site name="start_red" pos="{ -R1 / 2} 0 0" type="sphere"
size="0.001" rgba="0 1 0 0.5" />
    </body>

    <body name="green_mid" pos="{a + c / 2} 0 0">
      <site name="mid_green" pos="0 0 0" type="sphere" size="0.001"/>
      <joint name="mid_joint_x_green" type="slide" axis="1 0 0"/>
      <joint name="mid_joint_y_green" type="slide" axis="0 0 1"/>
      <geom type="sphere" size="0.001" mass="0.0001" rgba="0 1 0 0.5"
contype="0"/>
    </body>

    <body name="red_mid" pos="{a + c / 2} 0 0">
      <site name="mid_red" pos="0 0 0" type="sphere" size="0.001"/>
      <joint name="mid_joint_x_red" type="slide" axis="1 0 0"/>
      <joint name="mid_joint_y_red" type="slide" axis="0 0 1"/>
      <geom type="sphere" size="0.001" mass="0.0001" rgba="1 0 0 0.5"
contype="0"/>
    </body>

    <body name="link_end" pos="{a + b + c} 0 0">
      <site name="end_link" pos="0 0 0" type="sphere" size="0.001"/>
      <joint name="end_x" type="slide" axis="1 0 0"/>
      <joint name="end_y" type="slide" axis="0 0 1"/>
      <geom type="box" size="0.002 0.002 {R2 / 2}" rgba="1 1 0 0.5"
mass="0.001" contype="0"/>
    </body>

    <body name="link_1" pos="0 0 0">

```

```

        <geom type="cylinder" pos="{a / 2} 0 0" size="0.0002 {a / 2}"
euler="0 90 0" rgba="0 1 1 0.5" contype="0"/>

        <body name="link_2" pos="{a} 0 0">
            <joint name="rotate_1" type="hinge" axis="0 1 0"
stiffness="0" springref="0" damping="0"/>
            <geom type="cylinder" pos="{c / 2} 0 0" size="0.0002 {c / 2}"
euler="0 90 0" rgba="0 1 1 0.5" contype="0"/>

            <geom name="pulley_r1" type="cylinder" size="{R1 / 2} 0.01"
pos="0 0 0" euler="90 0 0" rgba="1 0 0 0.5" contype="0"/>
            <site name="side_r1_green" pos="0 0 {-R1 / 2 - 0.0001}"
type="sphere" size="0.001"/>
            <site name="side_r1_red" pos="0 0 {R1 / 2 + 0.0001}"
type="sphere" size="0.001"/>
            <site name="pulley_r1_center" pos="0 0 0" type="sphere"
size="0.001"/>

            <body name="link_3" pos="{c} 0 0">
                <joint name="rotate_2" type="hinge" axis="0 1 0"
stiffness="0" springref="0" damping="0"/>
                <geom type="cylinder" pos="{b / 2} 0 0" size="0.0002 {b /
2}" euler="0 90 0" rgba="0 1 1 0.5" contype="0"/>

                <geom name="pulley_r2" type="cylinder" size="{R2 / 2}
0.01" pos="0 0 0" euler="90 0 0" rgba="0 1 0 0.5" contype="0"/>
                <site name="side_r2_green" pos="0 0 {R2 / 2 + 0.0001}"
type="sphere" size="0.001"/>
                <site name="side_r2_red" pos="0 0 {-R2 / 2 - 0.0001}"
type="sphere" size="0.001"/>
                <site name="pulley_r2_center" pos="0 0 0" type="sphere"
size="0.001"/>

                <site name="end" pos="{b} 0 0" type="sphere"
size="0.001"/>

            </body>
        </body>
    </body>
</worldbody>

    <tendon>
        <spatial name="tendon_green" width="0.001" rgba="1 0 0 0.5"
stiffness="1" damping="10" springlength="0.005">
            <site site="start_green"/>
            <geom geom="pulley_r1" sidesite="side_r1_green"/>

```

```

        <site site="mid_green"/>
        <geom geom="pulley_r2" sidesite="side_r2_green"/>
        <site site="end"/>
    </spatial>

    <spatial name="tendon_red" width="0.001" rgba="0 1 0 0.5"
stiffness="1" damping="10" springlength="0.005">
        <site site="start_red"/>
        <geom geom="pulley_r1" sidesite="side_r1_red"/>
        <site site="mid_red"/>
        <geom geom="pulley_r2" sidesite="side_r2_red"/>
        <site site="end"/>
    </spatial>
</tendon>

<equality>
    <weld site1="end" site2="end_link" torquescale="100"/>

    <connect site1="mid_green" site2="pulley_r1_center"/>
    <connect site1="mid_green" site2="pulley_r2_center"/>
    <connect site1="mid_red" site2="pulley_r1_center"/>
    <connect site1="mid_red" site2="pulley_r2_center"/>
</equality>

<actuator>
    <velocity name="motor_green" tendon="tendon_green" gear="1"
ctrlrange="-1 1"/>
    <velocity name="motor_red" tendon="tendon_red" gear="1" ctrlrange="-1
1"/>
</actuator>

</mujoco>
"""

```

```

def main():
    # Generate the XML model
    xml = generate_model_xml(R1, R2, a, b, c)
    # Load the model and create the data structure
    model = mujoco.MjModel.from_xml_string(xml)
    data = mujoco.MjData(model)
    # Set the test control signals for the actuators
    data.ctrl[0] = PULLEY_SPEED_1
    data.ctrl[1] = PULLEY_SPEED_2

    end_positions = []

```

```

try:
    with mujoco.viewer.launch_passive(model, data) as viewer:

        while viewer.is_running():
            # Simulation step loop
            simstart = data.time
            while data.time - simstart < 1.0/60.0:
                mujoco.mj_step(model, data)
                end_pos = data.site('end').xpos.copy()
                end_positions.append(end_pos)
            # Update the viewer's scene
            viewer.sync()
            # Sleep to maintain the real-time simulation rate
            time.sleep(model.opt.timestep - data.time % model.opt.timestep)

except Exception as e:
    # Handle potential errors in the simulation process
    print(f"Error during simulation: {e}")
    sys.exit(1)

# After simulation, plot the trajectory
end_positions = np.array(end_positions)
plt.figure()
plt.plot(end_positions[:, 0], end_positions[:, 2]) # Assuming planar in x-z
plt.xlabel('X position')
plt.ylabel('Z position')
plt.title('End Effector Trajectory')
plt.grid(True)
plt.savefig('end_effector_trajectory.png')
plt.show()

if __name__ == "__main__":
    main()

```